

Zhiheng Liu

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WORK EXPERIENCE

TikTok Pte. Ltd

Aug 2021 - Present

Software Engineer @ Multimedia Arch

Architected LLM-powered solutions and edge-cloud systems to optimize AV experiences & drive AI-driven efficiency.

LLM-Driven Health Check Systems

- Built LLM-based fault attribution tool, automating alarm analysis and reducing troubleshooting time by **40%+**; achieved reason tracing via Eino+Fornax integration.
- Launched LLM-powered experiment analysis workflow for **10+ daily report generation** requests.
- Developed prompt testing, mermaid generation web apps using Bolt for daily tool usage.

Edge-Cloud Collaboration Architecture

- Led 'Edge-Cloud Collaboration' platform server-side design, enabling bidirectional feature flows and end-to-end video profile application; drove QoE improvements, **Active Days +0.03%** etc. and **-6% cdn costs**.
- Engineered strategy orchestration system for real-time client execution (e.g., thread pool optimization); built multi-treatment platform solution to resolve experiment metric dilution, accelerating strategy validation.

Multimedia System Optimization

- Spearheaded storage optimization system, cutting per-video storage by 90% and **saving 500PB+** data in 1Q to meet 87% of annual cost targets; improved server **throughput 10x** via pprof-driven optimizations.
- Deployed P2P technology, reducing bandwidth costs by 0.05% in launched area;
- Implemented Upload Mini-Edge system to improve upload latency, and improved scheduling system with online stats.

Leadership & Impact

- Mentored 1 junior engineer into a P2P project lead; 1 intern in the LLM-integration project. Authored 10+ technical whitepapers with 2k+ reads.
- Collaborated across teams (client, algorithms, server) to standardize edge-cloud architecture and launch 4 major initiatives.

EDUCATION

Nanyang Technological University

Aug 2017 - June 2021

B.Eng. in Electrical and Electronic Engineering

SM2 Scholarship

Dean's List 2018 2019 2021

Highest Distinction, GPA 4.92/5.00